METALNET: DEPENDENCY-FREE C++ CNN HEADER-ONLY LIBRARY

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DECLARATION CERTIFICATE

This is to certify that the work presented in the project entitled **"MetalNet: Dependency-Free C++ CNN Header-Only Library"** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering at Cambridge Institute of Technology, Tatisilwai, Ranchi is an authentic work carried out under my supervision and guidance.

To the best of my knowledge, the content of this project does not form a basis for the award of any previous Degree to anyone else.

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CERTIFICATE OF APPROVAL

The foregoing project entitled "MetalNet: Dependency-Free C++ CNN Header-Only Library" is hereby approved as a creditable work on the topic and has been presented satisfactorily to warrant its acceptance as a prerequisite to the degree for which it has been submitted.

It is understood that by this approval, the undersigned does not necessarily endorse any conclusion drawn or opinion expressed therein, but approves the project for the purpose for which it is submitted.

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ABSTRACT

Deep learning vision systems typically rely on large frameworks like PyTorch or TensorFlow, which carry significant dependency overhead and are designed primarily for GPUs. On CPU-only environments such as laptops or edge devices, these frameworks introduce unnecessary resource consumption. This project presents **MetalNet**, a high-performance Convolutional Neural Network (CNN) library written from scratch in pure C++20. MetalNet is fully **header-only** and **dependency-free**, requiring no installation, linking, or configuration, and is designed specifically to optimize CPU execution.

Despite its lightweight design, MetalNet achieves high performance by treating the CPU as a first-class citizen. It implements an **Implicit GEMM** convolution algorithm that avoids memory-heavy *im2col* buffering, and stores data in the cache-friendly **NHWC** layout. Innermost arithmetic operations are accelerated using hand-written **AVX2 and FMA** SIMD instructions, while loop tiling and parallelization via **OpenMP** optimize multi-core CPU scaling. Together, these optimizations yield a **57× speedup** in convolution operations over a naive scalar baseline.

MetalNet supports both training and inference. It features a Directed Acyclic Graph (DAG) based automatic differentiation engine, core layers (Convolutional, Batch Normalization, Pooling, Dense), activation functions, SGD and Adam optimizers, and a built-in profiler. Tested against PyTorch on LeNet-5, VGG-style, and ResNet-18 architectures, MetalNet consistently runs **1.6–2.4× faster** for inference on the same CPU while reducing peak memory consumption by approximately **70%**. MetalNet demonstrates that high-performance deep learning is achievable on standard CPUs without complex external dependencies.

Keywords: Convolutional Neural Networks, C++20, Header-Only Library, SIMD Optimization, Implicit GEMM, Automatic Differentiation, AVX2, Cache Optimization, OpenMP, Deep Learning, CPU Performance

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LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
GEMM	General Matrix Multiplication
SIMD	Single Instruction, Multiple Data
AVX2	Advanced Vector Extensions 2
FMA	Fused Multiply–Add
im2col	Image to Column
DAG	Directed Acyclic Graph
SGD	Stochastic Gradient Descent
BN	Batch Normalization
ReLU	Rectified Linear Unit
NHWC	Batch, Height, Width, Channels
NCHW	Batch, Channels, Height, Width
L1/L2	Level 1 / Level 2 Cache
OpenMP	Open Multi-Processing
GPU	Graphics Processing Unit
CPU	Central Processing Unit
MNIST	Modified NIST handwritten digit dataset
CIFAR	Canadian Institute For Advanced Research
VGG	Visual Geometry Group
API	Application Programming Interface
DFD	Data Flow Diagram
UML	Unified Modelling Language
UAT	User Acceptance Testing
IDE	Integrated Development Environment